

Quispe & Xu (2026)

Agentic Delegation and the Language Frontier of Software Developers — the model, and what Lean says about it

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Repository 3 · Week of August 31 – September 5, 2026

`github.com/Arriola123456/ai-03-quispe`

Asked an LLM to summarise “Quispe (2026), *Coding Beyond Your Training: Claude Code and the Technological Frontier of Software Developers*.” Then opened the arXiv record.

	Citation as given	arXiv:2605.25438
Title	<i>Coding Beyond Your Training: Claude Code and the Technological Frontier of Software Developers</i>	<i>Agentic Delegation and the Language Frontier of Software Developers: A Model and Evidence from Claude Code on GitHub</i>
Authors	Quispe	Quispe & Xu
Date	“2026”	v1 2026-05-25; v2 2026-07-07 (the Lean task pins v2)
Sample	—	5,346 developers; 57 million changed files; 28-month panel

The discrepancy that matters: “technological” vs. “language” frontier. Remark 1 of the paper insists the frontier is a *production* frontier, not a *skill* frontier — the wrong title erases the paper’s own caveat.

1 · The paper and the agent's problem

Question. Why does a Python-only developer ship Rust the month she adopts an *agent*, when two years of ChatGPT/Copilot did nothing of the kind?

Primitives at a fixed (i, k, t) . Opportunity ω , activation cost b , language skill $s \in [0, 1]$, beliefs $\theta \sim N(\mu, 1/\pi)$, general ability a , CARA utility $u(y) = -e^{-\rho y}$ so that $CE = m - \rho\sigma^2/2$.

Three production modes (certainty-equivalent surpluses):

$$\begin{aligned} V^S &= \omega + s\mu - \frac{\rho s^2}{2\pi} - b, & V^C &= V^S + \gamma s - r^C, \\ V^D &= \omega + (1 - \lambda)s\mu + \lambda az(A) - \kappa(a, s) - r^D - b - \frac{\rho}{2} \left[\frac{(1 - \lambda)^2 s^2}{\pi} + \sigma_D^2(a, s, A) \right]. \end{aligned}$$

Menus $M_1 = \{S, C\}$ (before) and $M_2 = \{S, C, D\}$ (after); $V^g = \max_{m \in M_g} V^m$; language active iff $Z^g = 1[V^g \geq 0]$; $N_{it}^g = \sum_k Z_{ik,t}^g$.

Assumption 1. Unfamiliar: $\gamma s - r^C \leq 0$; familiar: $\gamma \bar{s} - r^C > 0$. **Assumption 2.** $\kappa_a < 0$, $\kappa_s \leq 0$, $\partial \sigma_D^2 / \partial a, \partial s, \partial A \leq 0$.

2 · Thresholds and the activation band

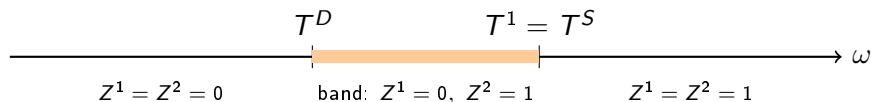
Every surplus is affine in ω with slope one, so $V^m = \omega - T^m$:

$$T^S = b - s\mu + \frac{\rho s^2}{2\pi}, \quad T^C = T^S - (\gamma s - r^C), \quad T^1 = \min\{T^S, T^C\} = T^S - \max\{0, \gamma s - r^C\}.$$

$$T^D = b - (1 - \lambda)s\mu - \lambda az(A) + \kappa(a, s) + r^D + \frac{\rho}{2} \left[\frac{(1 - \lambda)^2 s^2}{\pi} + \sigma_D^2 \right], \quad T^2 = \min\{T^1, T^D\} \leq T^1.$$

For an unfamiliar language Assumption 1 gives $T^1 = T^S$, and the **agentic threshold reduction** is

$$B \equiv T^1 - T^D = \lambda[az(A) - s\mu] - \kappa(a, s) - r^D + \frac{\rho}{2} \left[\frac{(2\lambda - \lambda^2)s^2}{\pi} - \sigma_D^2(a, s, A) \right].$$



3 · The main result, with all its conditions

Proposition 2 (Activation band). *Consider an unfamiliar language satisfying Assumption 1. If $B_{ik,t} > 0$, then*

$$Z_{ik,t}^2 - Z_{ik,t}^1 = 1 [T_{ik,t}^D \leq \omega_{ik,t} < T_{ik,t}^S].$$

- (i) **Unfamiliar language**, so Assumption 1 applies: $\gamma^S - r^C \leq 0 \Rightarrow T^1 = T^S$.
- (ii) $B > 0$, i.e. $T^D < T^S$: the band is nonempty.
- (iii) CARA–Normal certainty equivalents behind V^S, V^C, V^D (Appendix A.1); $\rho > 0, \pi > 0, \lambda \in (0, 1]$.
- (iv) Menu choice is max over modes; Z^g is the indicator of nonnegative surplus.
- (v) With a continuous conditional CDF F : $\Pr(\text{band}) = F(T^S) - F(T^D)$ and $E[N^2 - N^1] = \sum_k [F(T_k^1) - F(T_k^2)] \geq 0$ (Eq. 9).

What it does not claim (Remark 1): $s_{ik,t}$ does not rise at adoption; the developer ships Rust by *directing* an agent. Proposition 1 ($Z^2 \geq Z^1$) is “nearly mechanical” — a larger menu.

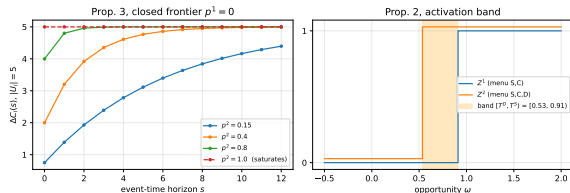
What Lean added: the identity holds *without* $B > 0$ (empty band, both sides 0); the premise only buys nonemptiness.

4 · Proposition 3 and the endpoint of its domain

Per-period first-use hazard p_{ik}^g ; cumulative effect at horizon s :

$$\Delta C_i(s) = \sum_{k \in U_i} \left[(1 - p_{ik}^1)^{s+1} - (1 - p_{ik}^2)^{s+1} \right] \geq 0 \quad \text{if } p_{ik}^2 \geq p_{ik}^1,$$

“which in the closed-frontier benchmark $p_{ik}^1 = 0 < p_{ik}^2$ is **strictly increasing and concave.**” Appendix A.6: first difference $p^2(1 - p^2)^{s+1} > 0$, second $-(p^2)^2(1 - p^2)^{s+1} < 0$.



At $p^2 = 1$, which the printed domain allows: $\Delta C_i(s) = |U_i|$ for every s — constant, not strictly increasing; both displayed inequalities are equalities.

Corrected reading: strict increase and strict concavity for $0 < p^2 < 1$; weak monotonicity on the whole domain. Lean states the strict rows with $p^2 < 1$ and proves the $p^2 = 1$ saturation separately.

5 · Two papers, one phenomenon, one modelling choice

Aouad–Lykouris–Zhong (week 1)

AI is a *perfectly substitutable input* on the *intensive* margin of a task the worker already does:

$$x = s + e + a, \quad \max_{e \geq 0} p(s + e + a) - \gamma e.$$

Assistance crowds out effort; effort builds skill $\Rightarrow$ **deskilling**.

The single choice. Whether AI enters *inside* the production function of an existing task (a substitute for e , so it displaces the thing that builds s) or *beside* it as an alternative technology for a task that was infeasible. Neither paper needs the other's mechanism to be false; they model different margins. A model with both would have deskilling in familiar languages and expansion in unfamiliar ones — which is, roughly, the paper's own Figure 1 read together with Remark 1.

Quispe–Xu

Agentic AI is a *new production mode* on the *extensive* margin, with its own threshold that does not load on s :

$$V^2 = \max\{V^S, V^C, V^D\} \geq V^1.$$

It adds options instead of replacing inputs
 $\Rightarrow$ **frontier expansion**.

6 · What I did analytically

Equivalente Certo:

Para $Y \sim N(m, \sigma^2)$ y $U(Y) = -e^{-\rho Y}$ se utiliza su moment generating function de una normal (MGF), tal que:

$$\mathbb{E}[e^{tY}] = e^{tm + \frac{1}{2}t^2\sigma^2}$$

Ahora, nosotros buscamos el equivalente certo CE, tal que:

$$U(CE) = \mathbb{E}[U(Y)]$$

Con la ecuación anterior calculamos el lado derecho de la igualdad:

$$\mathbb{E}[U(Y)] = \mathbb{E}[-e^{-\rho Y}] = -\mathbb{E}[e^{-\rho Y}]$$

Sustituyendo $t = -\rho$:

$$-\mathbb{E}[e^{-\rho Y}] = -\exp\left[(-\rho)m + \frac{1}{2}(-\rho)^2\sigma^2\right]$$

$$\mathbb{E}[U(Y)] = -\exp\left[-\rho m + \frac{\rho^2\sigma^2}{2}\right]$$

Ahora, se calcula el CE:

$$U(CE) = -e^{-\rho CE} = \exp[-\rho CE] = -\exp\left[-\rho m + \frac{\rho^2\sigma^2}{2}\right] = \mathbb{E}[U(Y)]$$

$$-\rho CE = -\rho m + \frac{\rho^2\sigma^2}{2}$$

$$CE = m - \frac{\rho\sigma^2}{2}$$

$$CE = \underbrace{m}_{\text{Payoff Esperado}} - \underbrace{\frac{\rho\sigma^2}{2}}_{\text{Risk Penalty}}$$

By hand (hand/, 5 pages). The step I did not want to take on trust: the CARA–Normal certainty equivalent. With $Y \sim N(m, \sigma^2)$ and $u = -e^{-\rho Y}$, the MGF gives $\mathbb{E}[-e^{-\rho Y}] = -\exp(-\rho m + \rho^2\sigma^2/2)$, hence $CE = m - \rho\sigma^2/2$. Then V^S from $Y^S = \omega + s\theta - b$ ($\text{Var} = s^2/\pi$), V^C , and V^D with $\text{Var}[Y^D] = (1 - \lambda)^2 s^2/\pi + \sigma_D^2$.

Threshold algebra. $T^S - T^D$ reduces to Equation (7) because $\frac{\rho}{2} \frac{s^2}{\pi} [1 - (1 - \lambda)^2] = \frac{\rho}{2} (2\lambda - \lambda^2) \frac{s^2}{\pi}$.

Endpoint checks. $\lambda = 1$: fine (human match risk vanishes). $\pi \rightarrow \infty$: fine. $B \leq 0$: identity (8) still holds, band empty. $\rho^2 = 1$: Proposition 3's strict claims fail. Comparative static “ B larger for higher a ” needs $z(A) \geq 0$ and $\rho \geq 0$, which the text leaves implicit.

7 · What I did computationally — and with which tools

EconCSLib workflow (nikhgarg/EconCSLib, commit cf500b74, Lean/Mathlib v4.30.0-rc2, WSL2 Ubuntu 26.04):

1. pin the v2 PDF outside the repo (SHA-256 cddc0487...3d35); `init-spec`;
2. fill the **statement spec**: 15 targets, each with source locator, literal source statement and a transparent Lean proposition;
3. `paper_contribution.py new` → scaffold (Lean-validates every Spec);
4. `MainTheorems.lean` (model, threshold algebra) + `ProofInterface.lean` (15 endpoints);
5. `lake build QX26AgenticDelegation`: **Build completed successfully (4000 jobs)**; check `-fast`: **exit 0**; full check: build and status lanes pass, stops at the conclusion-provenance audit (needs the LLM-as-judge sidecars; `lean/docs/CHECK_FULL_OUTPUT.txt`).

Result. 15/15 Specs proved, no sorry; status *partially formalized* (two Proposition 3 rows carry the added $p^2 < 1$; semantic audits pending).

Who ran what — the blocker. Codex gpt-5.6-sol/xhigh was launched exactly as required; it read the skills, wrote the placeholder spec, tested a scratch file (parse errors: $\Rightarrow$ for $\mapsto$) and died: `usage_limit_exceeded`, “Your workspace is out of credits.” Steps 2–5 were done with Claude Code — disclosed in `prompts.md` and `lean/docs/RUN_LOG.md`; nothing in `lean/` is presented as the Codex run.

8 · Lean: Proposition 3 — from the equation to a checked proof

Paper claim (Prop. 3, A.6). For $p^1 = 0 < p^2$, $\Delta C(s) = \sum_{k \in U} [(1 - 0)^{s+1} - (1 - p_k^2)^{s+1}]$ is strictly increasing in s .

Lean statement (PaperInterface.lean, transparent Spec):

```
def paper_proposition_3_closed_frontier_strictly_increasingSpec : Prop :=
  ∀ {ι : Type} (U : Finset ι) (p2 : ι → ℝ) (s : ℕ),
    U.Nonempty → (∀ k, 0 < p2 k) → (∀ k, p2 k < 1) →
      ∑ k ∈ U, ((1 - (0 : ℝ)) ^ (s + 1) - (1 - p2 k) ^ (s + 1))
        < ∑ k ∈ U, ((1 - (0 : ℝ)) ^ (s + 2) - (1 - p2 k) ^ (s + 2))
```

Proof fragment (ProofInterface.lean):

```
apply Finset.sum_lt_sum_of_nonempty hU -- needs U ≠ ∅
intro k _
have hq0 : 0 < 1 - p2 k := by linarith [h1 k] -- needs p2 k < 1 (added)
have hq1 : 1 - p2 k < 1 := by linarith [h0 k] -- needs 0 < p2 k
have := pow_lt_pow_right_of_lt_one0 hq0 hq1 (show s + 1 < s + 2 by omega)
simp only [sub_zero, one_pow]; linarith
```

Reading. $U_i \mapsto \text{Finset}$, hazards $\mapsto p2 : \iota \rightarrow \mathbb{R}$, horizon $\mapsto s : \mathbb{N}$; the strict step $\text{pow_lt_pow_right_of_lt_one0}$ needs $0 < 1 - p < 1$.

The endpoint theorem (MainTheorems.lean):

```

theorem cumulative_effect_saturates_at_hazard_one {ι : Type} (U : Finset ι)
  (p2 : ι → ℝ) (s : ℕ) (h : ∀ k, p2 k = 1) :
  ∑ k ∈ U, ((1 - (0 : ℝ)) ^ (s + 1) - (1 - p2 k) ^ (s + 1))
  = ∑ k ∈ U, ((1 - (0 : ℝ)) ^ (s + 2) - (1 - p2 k) ^ (s + 2)) := by
  apply Finset.sum_congr rfl; intro k _; rw [h k]; simp

```

- ▶ **Objects.** $U_i \mapsto U : \text{Finset } \iota$; $p_{ik}^2 \mapsto p2\ k$; the horizon $s \in \mathbb{N}$; $\Delta C_i(s)$ is written out as the source sum with $p^1 = 0$ kept explicit as $(1 - (0 : \mathbb{R}))$.
- ▶ **Assumptions.** The source says only $p^1 = 0 < p^2$. Lean would not close the strict inequality with that: `pow_lt_pow_right_of_lt_one0` demands $1 - p^2 < 1$ and $0 < 1 - p^2$, i.e. $0 < p^2 < 1$; the sum lemma demands $U \neq \emptyset$.
- ▶ **Conclusion.** Lean verifies the source claim **only after adding** $p^2 < 1$ and $U \neq \emptyset$; the second theorem proves that at $p^2 = 1$ the two sides are *equal*, so the printed strict claim is false on its stated domain.
- ▶ **Check.** `lake build QX26AgenticDelegation`: Build completed successfully (4000 jobs); `paper_contribution.py check -fast`: exit 0; no sorry.

Paper claim. Assumption 1 and $B > 0 \Rightarrow Z^2 - Z^1 = 1[T^D \leq \omega < T^S]$. (lam = λ , prec = π , z = $z(A)$, κ , σ at fixed (i, k, t))

```
def paper_proposition_2_activation_bandSpec : Prop :=
  ∀ ω s μ prec b ρ γ rC lam a z κ rD σ : ℝ,
    let VS := ω + s * μ - ρ * s ^ 2 / (2 * prec) - b
    let VC := VS + γ * s - rC
    let VD := ω + (1 - lam) * s * μ + lam * a * z - κ - rD - b
               - ρ / 2 * ((1 - lam) ^ 2 * s ^ 2 / prec + σ)
    let TS := b - s * μ + ρ * s ^ 2 / (2 * prec)
    let TD := b - (1 - lam) * s * μ - lam * a * z + κ + rD
               + ρ / 2 * ((1 - lam) ^ 2 * s ^ 2 / prec + σ)
    γ * s - rC ≤ 0 → 0 < TS - TD →
      (if 0 ≤ max (max VS VC) VD then (1 : ℝ) else 0)
        - (if 0 ≤ max VS VC then (1 : ℝ) else 0)
        = if TD ≤ ω ∧ ω < TS then (1 : ℝ) else 0
```

Reading. Equations (1)–(3) are the lets; the menus are the nested max; Z^g is the if; Assumption 1 is $\gamma * s - rC \leq 0$; $B > 0$ is $0 < TS - TD$. Every quantifier and inequality of the source is visible — nothing hides behind a definition.

11 · Lean: Proposition 2 — the proof route

Route. $V^C \leq V^S$ (Assumption 1) $\Rightarrow \max VS VC = VS$; rewrite $V^S = \omega - T^S$, $V^D = \omega - T^D$ (ring); then one real lemma:

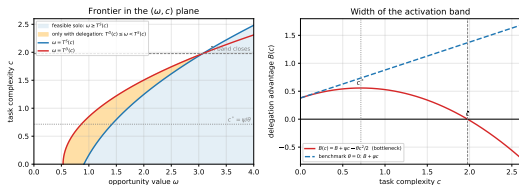
```
theorem band_indicator {ω TS TD : ℝ} (hlt : TD < TS) :
  (if 0 ≤ max (ω - TS) (ω - TD) then (1:ℝ) else 0) - (if 0 ≤ ω - TS then (1:ℝ) else 0)
    = if TD ≤ ω ∧ ω < TS then (1:ℝ) else 0 := by
have hmax : max (ω - TS) (ω - TD) = ω - TD := max_eq_right (by linarith)
rw [hmax]
by_cases hS : TS ≤ ω          -- case 1: ω ≥ TS → 1 - 1 = 0, band indicator 0
· ...                        -- case 2: TD ≤ ω < TS → 1 - 0 = 1, band indicator 1
· by_cases hD : TD ≤ ω      -- case 3: ω < TD → 0 - 0 = 0
  · ...
```

- ▶ Each case is closed by `if_pos/if_neg` on the three conditions and `norm_num` on the resulting $1 - 1$, $1 - 0$, $0 - 0$.
- ▶ Lean verifies the source claim **exactly**: no added hypothesis, no changed domain.
- ▶ **Extension** `activation_band_general` drops $0 < TS - TD$: when $T^D \geq T^S$ the band is empty and both sides are 0 — the premise $B > 0$ only makes the band nonempty.

12 · Extension: task complexity and the verification bottleneck

Idea (from my ChatGPT study session). Complexity $c \geq 0$ raises the project's value $R(c)$ and its execution cost in both modes — and, when delegated, the cost of *verifying* the agent. $R(c)$ and the common curvature χ cancel in B :

$$T^S(c) = T^S + \frac{\chi}{2}c^2 + \psi c, \quad T^D(c) = T^D + \frac{\chi}{2}c^2 + \frac{\eta}{2}c^2 + \frac{\rho}{2}\frac{\nu}{2}c^2 \implies B(c) = B + \psi c - \frac{\theta}{2}c^2, \quad \theta \equiv \eta + \frac{\rho\nu}{2}.$$



Benchmark ($\theta = 0$): $B + \psi c$ grows without bound — the conclusion is the assumption.

Bottleneck ($\theta > 0$), proved in Lean

(`MainTheorems.lean`): peak at $c^* = \psi/\theta$ since $B(c^*) - B(c) = \frac{\theta}{2}(c - c^*)^2$

(`reductionC_le_peak`); strictly increasing below and decreasing above c^* ; the band *closes*,

$B(c) < 0$ beyond an explicit $\bar{c}$

(`reductionC_neg_of_large`); c^* rises with ability when η, ν fall with a (Assumption 2's logic).

Prediction. Newly-used languages concentrate in projects of *intermediate* complexity — testable with commit size.

13 · Where I did not believe the AI

Entonces, el payoff de Y^D es:

$$\mathbb{E}[Y^D] = W + (1-\lambda)s\theta + \lambda a_2(A) - K(a_1S) - r_0 - b$$

$$Y^D = W + (1-\lambda)s\theta + \lambda a_2(A) - K(a_1S) - r_0 - b$$

Se calcula la varianza:

$$\text{Var}[Y^D] = \text{Var}[(1-\lambda)s\theta] = (1-\lambda)^2 s^2 \text{Var}(\theta) = (1-\lambda)^2 s^2 \frac{1}{\pi}$$

Ahora el paper introduce $\sigma_0^2(a_1, s, A)$, pues Claude pudo introducir un bug y entonces:

$$\text{Var}[Y^D] = (1-\lambda)^2 \frac{s^2}{\pi} + \sigma_0^2(a_1, s, A)$$

Ahora se usa la ecuación $CE = m - \frac{\rho}{2}\sigma^2$, tal que

$$m = W + (1-\lambda)s\theta + \lambda a_2(A) - K(a_1S) - r_0 - b = \mathbb{E}[Y^D]$$

$$\sigma^2 = (1-\lambda)^2 \frac{s^2}{\pi} + \sigma_0^2(a_1, s, A) = \text{Var}[Y^D]$$

Entonces:

$$V^D = \underbrace{W}_{\text{Proyecto}} + \underbrace{(1-\lambda)s\theta}_{\text{Payoff Humano}} + \underbrace{\lambda a_2(A)}_{\text{Payoff AI+to}} - \underbrace{K(a_1S)}_{\text{Costos}} - \underbrace{r_0 - b}_{\text{Reserva}} - \frac{\rho}{2} \left[\underbrace{(1-\lambda)^2 \frac{s^2}{\pi}}_{\text{Riesgo}} + \sigma_0^2 \right]$$

Claim 1 (an LLM, cold). A confident summary of “*Coding Beyond Your Training*” — a title that does not exist, one author instead of two. **Verdict:** fabricated (slide 0).

Claim 2 (the risk term in V^D). Delegation shrinks the human-match variance to $(1-\lambda)^2 s^2 / \pi$ and adds σ_D^2 ; the penalty is $\frac{\rho}{2}$ times the variance. Checked by hand, on screen: MGF $\rightarrow CE = m - \rho\sigma^2/2 \rightarrow \text{Var}[(1-\lambda)s\theta] = (1-\lambda)^2 s^2 / \pi$. **Verdict:** Equation (3) is right as printed.

Claim 3 (Codex’s draft Lean). Wrote $\Rightarrow$ where Lean 4 wants $\mapsto / \Rightarrow$; never reached a statement. **Verdict:** syntax, not mathematics — hence the scratch-file test of every Spec before scaffolding.

Claim 4 (ChatGPT, on the paper). Asked whether Proposition 3 has an error: “no error in the main logic; a small technical omission — strict monotonicity needs $0 < p^2 < 1$.” **Verdict:** right, and understated — at $p^2 = 1$ the printed claim is false, not merely incomplete; Lean proves the saturation (slide 9).

14 · This week's trap: where the argument is weakest

- ▶ **Does the mechanism need an *agent*?** In the model, no. What creates the band is that mode D 's value *does not load on* s while mode C 's value is γs (Assumption 1). Any assistance worth $\gamma' - r$ with γ' independent of s would lower T^1 for unfamiliar languages just as well. “Agent” is a *restriction on the shape of the assistance technology*, not a derived necessity. The paper's defence is Section 3 — the generational boundary is datable, so the shape assumption is testable — an empirical argument, not a theorem.
- ▶ **Endpoints.** $p^2 = 1$ breaks Proposition 3's strict claims (proved). $B > 0$ is not needed for Proposition 2's identity (proved). The comparative static after (7) needs $z(A) \geq 0$, $\rho \geq 0$ (added in Lean). Nothing breaks at $\lambda = 1$.
- ▶ **Representation gaps Lean makes visible.** Propositions 4 and 5 are stated over abstract hazards $p_i(a_i, A)$ and entry costs $c_r(\ell, g)$; the link from $Z^2 - Z^1$ to those objects is assumed, exactly as in the source proofs. Verification certifies the *algebra*, not the *economics* of the reduced form.
- ▶ **Selection into adoption** (empirical; flagged by the authors): the event-time estimates are associations. The model cannot distinguish a lower T^D from a project shock that raises ω at adoption — both put ω in the band.

15 · What is in the repository

`github.com/Arriola123456/ai-03-quispe`

- ▶ `README.md` — question, agent's problem, Proposition 2 with all its conditions, the Aouad contrast.
- ▶ `prompts.md` — raw, three sessions: the ChatGPT study session (the model term by term, Propositions 1–3, a complexity extension), the Codex session (107 tool calls, then `usage_limit_exceeded`) and the Claude Code session, including every Lean error hit and fixed.
- ▶ `hand/derivacion-a-mano.pdf` — CE via the MGF; V^S , V^C , V^D .
- ▶ `lean/` — the EconCSLib folder as generated: 15 Specs, 15 closed proofs, validation report, dependency DAG, audit stubs, `docs/RUN_LOG.md`, `docs/CHECK_FULL_OUTPUT.txt`.
- ▶ `analysis/` — the numerical checks behind slides 4 and 12.

One sentence. The model is right about the band and wrong about one endpoint; an LLM noticed the endpoint, and Lean would not let “ $0 < p^2$ ” stand in for “ $0 < p^2 < 1$ ”.